Gayathri Nutalapati

Data Engineer

5134005286 gayathrinutala@gmail.com

ABOUT ME

Highly qualified Data Engineer with 5 years of experience in the industry. Enjoy creative problem solving and getting exposure on multiple projects, and I would excel in the collaborative environment on which your company prides itself.

PROFILE SUMMARY

- Data engineer with **5** years of experience architecting and implementing end-to-end data pipelines, **ETL processes**, and analytics solutions on cloud platforms like **AWS, Azure,** and **on-premises Hadoop.**
- Practical knowledge in using **Sqoop** to import and export **data** from **Relational Database Systems** to **HDFS**, and vice versa
- Hands-on experience developing **data** pipelines using **Spark** components, **Spark SQL**, **Spark** Streaming and **MLLib**.
- Designed, build and managed **ELT data** pipelines leveraging **Airflow**, **python**, and **GCP** solutions.
- Has experience with story grooming, Sprint planning, daily standups, and software techniques like **Agile** and **SAFe**.
- Experienced with **Dimensional modelling, Data migration, Data cleansing, Data profiling,** and **ETL** Processes features for data warehouses. Worked with **csv**, **Avro**, **Parquet data** to load into **Data** frames and do the analysis.
- Experience in **Data extraction** (extract, Schemas, corrupt record handling and parallelized code), transformations and loads (user - defined functions, join optimizations) and Production (optimize and automate Extract, Transform and Load).
- Expert in designing Parallel jobs using various stages like **Join, Merge, Lookup, remove duplicates, Filter**, **Dataset**, **Lookup file set, Complex flat file, Modify, Aggregator, XML**. Good knowledge in **Database Creation** and maintenance of physical **data** models with **Oracle**, **Teradata**, Netezza, **DB2**, MongoDB, **HBase** and **SQL Server** databases.
- Experience in migrating on premise to Windows **Azure** using **Azure Site Recovery** and **Azure** backups.
- Worked with **Matillion** which Leverage **Snowflake**'s separate compute and storage resources for rapid transformation and get the Get the most from **Snowflake**-specific features, such as Alter Warehouse and Flatten Variant, Object, and Array.
- Experience in **Performance Monitoring**, **Security**, **Trouble shooting**, **Backup**, **Disaster recovery**, **Maintenance** and Support of **Linux** systems. Used **Spark** streaming, **HBase**, and **Kafka** to work on real-time **data** integration.
- Proficient with Container systems like **Docker** and Container orchestration like **EC2** Container Service, **Kubernetes**, worked with Terraform. Practical experience with **Python** and **Apache Airflow** to create, schedule, and monitor workflows
- Expertise in building **CI/CD** on **AWS** environment using **AWS** Code Commit, Code Build, Code Deploy and Code Pipeline and experience in using **AWS CloudFormation**, **API Gateway**, and **AWS Lambda** in automation and securing the infrastructure on **AWS**. Develop batch processing solutions by using **Data Factory** and **Azure Data** bricks.
- Experience in **Azure** Marketplace where to search, deploy and purchase wide range of applications and services.
- Extensive experience in relational **Data modeling**, **Dimensional data modeling**, **logical/Physical Design**, **ER Diagrams** and **OLTP** and **OLAP System Study** and **Analysis**.

TECHNICAL SKILLS

AWS Ecosystem	S3Bucket, Athena, Glue, EMR, Redshift, Data Lake, AWS Lambda, Kinesis
Azure Ecosystem	Azure Data Lake, ADF, Databricks, Azure SQL
Programming and Scripting	Python, Pyspark, SQL, HiveQL, Shell Scripting, Scala, HTML
Big Data Ecosystem	HDFS, Nifi, Map Reduce, Oozie, Hive/Impala, Pig, Sqoop, Zookeeper and HBase, Spark, Scala, Kafka, Apache Flink, Yarn, Cassandra, Cloudera, Horton works
Database Management	Snowflake, Teradata, Redshift, MySQL, SQL Server, Oracle, Dynamo DB.
Operating Systems	Windows, Unix, Linux
Reporting and ETL Tools	Tableau, PowerBI, Informatica, Airflow.

PROFESSIONAL EXPERIENCE

Client: Cardinal Health, Dublin, Ohio, USA **Jun 2023 to Present**

Role: Azure Data Engineer

Description: Cardinal Health is a distributor of pharmaceuticals, a global manufacturer and distributor of medical and laboratory products. I contribute to traditional data management systems as well as modern ones like Hadoop, Spark, Kafka, etc.

Responsibilities:

- Experience in using different types of stages like **Transformer, Aggregator, Merge, Join, Lookup, and Sort, remove duplicated, Funnel, Filter, Pivot for developing jobs**.
- Conducted performance tuning and optimization of **Kubernetes** and **Docker** deployments to improve overall system performance. Build **Data** pipelines using **Python**, **Apache Airflow** for **ETL** related jobs **inserting data** into **Oracle**.
- Presented the project to faculty and industry experts, showcasing the pipeline's effectiveness in providing real-time insights for marketing and brand management.
- Used **Python** based GUI components for the Front-End functionality such as selection criteria.
- Storing different configs in **No SQL database Mongo DB** and manipulating the configs using **PyMongo**.
- Used **Azure Data** factory to ingest **data** from log files and business custom applications, **processed data** on **Data bricks** per day-to-day requirements, and loaded them to **Azure Data Lake**s.
- Have worked on partition of **Kafka** messages and setting up the replication factors in **Kafka Cluster**.
- Experience in creating **Kubernetes** replication controllers, **Cluster**s and label services to deployed Microservices in **Docker**.
- Involved in the entire lifecycle of the projects including Design, Development, and Deployment, Testing and Implementation, and support. Performed Unit, System Integration Testing.
- Led requirement gathering, business analysis, and technical design for Hadoop and Big **Data** projects.
- Spearheaded **HBase** setup and utilized **Spark** and **SparkSQL** to develop faster **data** pipelines, resulting in a 60% reduction in processing time and improved **data accuracy**.
- Instantiated, created, and maintained **CI/CD** (continuous integration & deployment) pipelines and apply automation to environments and applications. Worked on various automation tools like **GIT, Terraform, Ansible**.
- Handled importing **data** from various **Data** sources, performed transformations using **Pandas**, **Spark** and loaded into **Hive** as external files using **HDFS**.
- Implemented Navigation rules for the application and page outcomes, written controllers using annotations.
- Implemented RESTful Web-**Services** for sending and receiving **data** between multiple systems.
- Successfully completed a POC for **Azure** implementation, with the larger goal of migrating on-premises servers and data to the cloud.
- Enhanced by adding **Python XML SOAP** request/response handlers to add accounts, modify trades and security updates.
- Created **Data stage** and **ETL** jobs for populating the data into Data Warehouse constantly from different source systems like ODS, flat files, Parquet. Created clusters to classify control and test groups.
- Developed multiple notebooks using **Pyspark** and **Spark SQL** in Databricks for data extraction, analyzing and transforming the data according to the business requirements.
- Utilized **Elasticsearch** and Kibana for indexing and visualizing the real-time analytics results, enabling stakeholders to gain actionable insights quickly.
- Used **Python**, R, **SQL** to create Statistical algorithms involving Multivariate Regression, Linea Regression, Logistic Worked on **Kafka** publishing the messages for further downstream systems. Designed and develop **JAVA API** (Commerce **API**) which provides functionality to connect to the **Cassandra** through **Java** services.
- Experience with **Microsoft Azure**, **Azure Resource Management templates, Virtual Networks, Storage, Virtual Machines**, and **Azure Active Directory**.
- Working on query languages such as **SQL**, code languages such as **Python** or **C#** and scripting languages such as PowerShell, M-Query (Power Query), or Windows batch commands.
- Worked with **Python ORM** classes with **SQL** Alchemy to get update the data from database.

Environment: Azure Data Factory, Apache Airflow, Talend, Spark, Azure SQL Database, Snowflake, SQL Server, Hive, T-SQL, Spark, Kafka, Azure Data Lake, Azure Synapse Analytics, Python, Scala, Tableau, Git.

Client: Progressive Insurance, Mayfield, Ohio, USA **Oct 2022 to May 2023**

Role: AWS Data Engineer

Description: The Progressive Corporation is an American insurance company. It became the largest motor insurance carrier in the USA. My responsibility is for optimizing data processing and query performance. This may involve tuning database queries, optimizing data storage, and leveraging distributed computing technologies.

Responsibilities:

- The **AWS Lambda** functions were written in **Spark** with cross - functional dependencies that generated custom libraries for delivering the **Lambda** function in the cloud. Performed raw data ingestion into, which triggered a lambda function and put refined data into ADLS.
- Integrated **Kubernetes** with cloud-native services, such as **AWS EKS** and **GCP GKE**, to leverage additional scalability and managed services. Collected and aggregated large amounts of web log data from different sources such as webservers, mobile and network devices using **Apache Flume** and stored the data into **HDFS** for analysis
- Working on data management disciplines including data integration, modeling and other areas directly relevant to business intelligence/business analytics development. Extracted and transformed the log data files from **S3** by Scheduling **AWS Glue** jobs and loaded the transformed data into Amazon Elastic search.
- Involved in development of **Web Services** using **SOAP** for sending and getting data from the external interface in the **XML** format.
- Used **Pig** as **ETL** tool to do Transformations with joins and pre-aggregations before storing the data onto **HDFS** and assisted Manager by providing automation strategies, **Selenium**/**Cucumber** Automation and **JIRA** reports.
- Ensured **data** quality and accuracy with custom **SQL** and **Hive** scripts and created data visualizations using **Python** and **Tableau** for improved insights and decision-making.
- Converted and parsed data formats using Py**Spark Data** Frames, reducing time spent on data conversion and parsing by 40%.
- Converted **SAS** codes to **Python** for predictive models with **Pandas**, **NumPy** and **scikit-learn**. Worked on **Java Message Service JMS API** for developing message-oriented middleware MOM layer for handling various asynchronous requests.
- Architected **Python** scripts for automated data extraction and loading from web server output files, reducing manual **data** entry, and processing time by 75%. Using **python NLTK** tool kit to be used in smart MMI interactions.
- Worked on creating **MapR**educe programs to parse the data for claim report generation and running the Jars in Hadoop. Co-ordinated with **Java** team in creating **MapReduce** programs.

Environment: Python, Pandas, NumPy, SSIS, PySpark, AWS, AWS S3, AWS DynamoDB, Amazon RDS, Amazon Redshift, AWS Glue, SQL Server, PostgreSQL, Tableau, Power BI, Apache Airflow.

Client: Deutsche Bank, Bangalore, India **Sep 2021 to Jul 2022**

Role: Application Developer/ Data Engineer

Description: Deutsche Bank is an Europe Large public sector bank. I involved in monitoring new and existing data pipelines to identify and resolve issues is an essential responsibility. It involves setting up monitoring systems and establishing procedures for quick troubleshooting and problem resolution.

Responsibilities:

- Conducted Performance tuning and optimization of **Snowflake data** warehouse, resulting in improved query execution times and reduced operational costs. Designed and deployed a **Kubernetes**-based containerized infrastructure for **data** processing and analytics, leading to a 20% increase in **data** processing capacity.
- Used **Python** to connect to **MySQL** using **MySQL** connectors and extracted various data for customer usage reports.
- Performed load testing and optimization to ensure the pipeline's scalability and efficiency in handling large volumes of data.
- Analyzed existing systems and propose improvements in processes and systems for usage of modern scheduling tools like **Airflow** and migrating the legacy systems into an Enterprise **data lake** built on **Azure Cloud**.
- Responsible for Building and Testing of applications. Experience in handling database issues and connections with **SQL** and **NoSQL** databases like MongoDB by installing and configuring various packages in **python** (**Teradata**, **MySQL**, **MySQL** connector, **PyMongo** and **SQL Alchemy**).
- Implemented Synapse Integration with **Azure Databricks** notebooks which reduce about half of development work. And achieved performance improvement on Synapse loading by implementing a dynamic partition switch.
- Staged the **API** and **Kafka Data** (in **JSON** file format) into **Snowflake** DB by Flattening the same for different functional services.

- Deployed models as **python** package, as **API** for backend integration and as services in a microservices architecture with a **Kubernetes** orchestration layer for the **Docker**s containers.
- Developed **JSON** Scripts for deploying the Pipeline in **Azure Data Factory** (ADF) that process the data using the **Sql** Activity and creating UNIX shell scripts for database connectivity and executing queries in parallel job execution.
- Consult leadership/stakeholders to share design recommendations and thoughts to identify product and technical requirements, resolve technical problems and suggest **Big Data** based analytical solutions.
- Developed **Spark** applications using **Scala** and spark **SQL** for data extraction, transformation, and aggregation from multiple file formats for analyzing and transforming the data uncover insight into the customer usage patterns and even.
- Build **Jenkins** jobs for **CI/CD** Infrastructure for **GitHub** repos.
- Responsible for loading the data from BDW **Oracle database**, **Teradata** into **HDFS** using **Sqoop**. Implemented AJAX, **JSON**, and **Java script** to create interactive web screens.
- Created Session Beans and controller Servlets for handling **HTTP** requests from Talend. Performed Data Visualization and Designed Dashboards with **Tableau** and generated complex reports including chars, summaries, and graphs to interpret the findings to the team and stakeholders.

Environment: Python, Django, JavaScript, MySQL, NumPy, SciPy, Pandas API, PEP, PIP, Jenkins, JSON, Git, JavaScript, AJAX, RESTful web service, MySQL, PyUnit.

Client: Myntra, Bangalore, India — Jun 2019 to Aug 2021
Role: Data Engineer

Description: Myntra is India's leading e-commerce store for fashion brand positioned at the sweet spot of aspiration and affordability. I implemented the processes and tools to ensure the accuracy, completeness, and consistency of data throughout its lifecycle. This includes data profiling, cleansing, and validation procedures.

Responsibilities:

- Maintain code version control and hold accountability for Prod-stage validation and deployment.
- Designed and implemented a fault-tolerant data processing framework leveraging **Kubernetes** and **Docker**, reducing downtime and increasing system reliability by 25%.
- Wrote and executed various **MYSQL** database queries from **Python** using **Python-MySQL** connector and **MySQL** db package.
- Created Pipelines in ADF using Linked **Services**/**Datasets**/Pipeline/ to Extract, Transform, and load data from different sources like **Azure SQL**, **Blob** storage, **Azure SQL Data** warehouse, write-back tool and backwards.
- Implemented airflow for workflow automation and scheduling tasks and created DAGs tasks.
- Involved in monitoring and scheduling the pipelines using Triggers in **Azure DataFactory**.
- Imported real time weblogs using **Kafka** as a messaging system and ingested the data to **Spark** Streaming and did data quality checks using **Spark** Streaming and arranged bad and passable flags on the data.
- Used Continuous Delivery Pipeline. Deployed microservices, including provisioning **Azure** environments and developed modules using **Python** scripting and Shell Scripting. Written queries in **MySQL** and **Native SQL**.
- Involved in various phases of Software Development Lifecycle (**SDLC**) of the application, like gathering requirements, design, development, deployment, and analysis of the application.
- Worked on **Big Data** Integration & **Analytics** based on **Hadoop**, **SOLR**, **PySpark**, **Kafka**, **Storm** and **webMethods**.
- Analyzed the **SQL** scripts and designed it by using **Spark SQL** for faster performance.
- Designed **GIT** branching strategies, merging per the needs of release frequency by implementing **GIT** flow workflow on **Bit bucket**.
- Involved in loading and transforming large sets of Structured, Semi-Structured and Unstructured data and analyzed them by running **Hive** queries. Processed the image data through the Hadoop distributed system by using Map and Reduce then stored into **HDFS**. Designing and implementing data integration solutions using **Azure Data Factory** to move data between various data sources, including on-premises and cloud-based systems.

Environment: Azure Data Factory, MS SQL, NoSQL, Azure HDInsight, SQL Server, PostgreSQL, PySpark, Azure SQL Data Warehouse, Azure Data Lake, Blob Storage.

EDUCATION

- Masters in Computer Science from University of Cincinnati, USA